

# Dataset Proposal — Phase 1

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## Dataset

UCI Bike Sharing Dataset — <https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset>

Hourly bike rental records from Washington D.C. (2011–2012). **17,379 rows, 17 columns.** Fully public, no personal data, used as a static CSV file. Features include a mix of numeric (temperature, humidity, windspeed) and categorical variables (season, hour, weekday, weather situation).

## Classification Question

*Will bike demand in a given hour be high or low?*

Target: a derived binary column `demand_level` — high if `cnt` is above the 70th percentile, low otherwise.

## Regression Question

*How many bikes will be rented in a given hour?*

Target: the existing `cnt` column (total hourly rentals) — no derivation needed.

## Why It Is Interesting

Predicting bike demand connects weather and time patterns to real operational decisions in urban transport — staffing, bike allocation, and station management. As an optional upgrade, the deployed app could pull **live weather data** from a free API (e.g. Open-Meteo) to generate real-time predictions, since weather is among the most predictive features in the dataset.